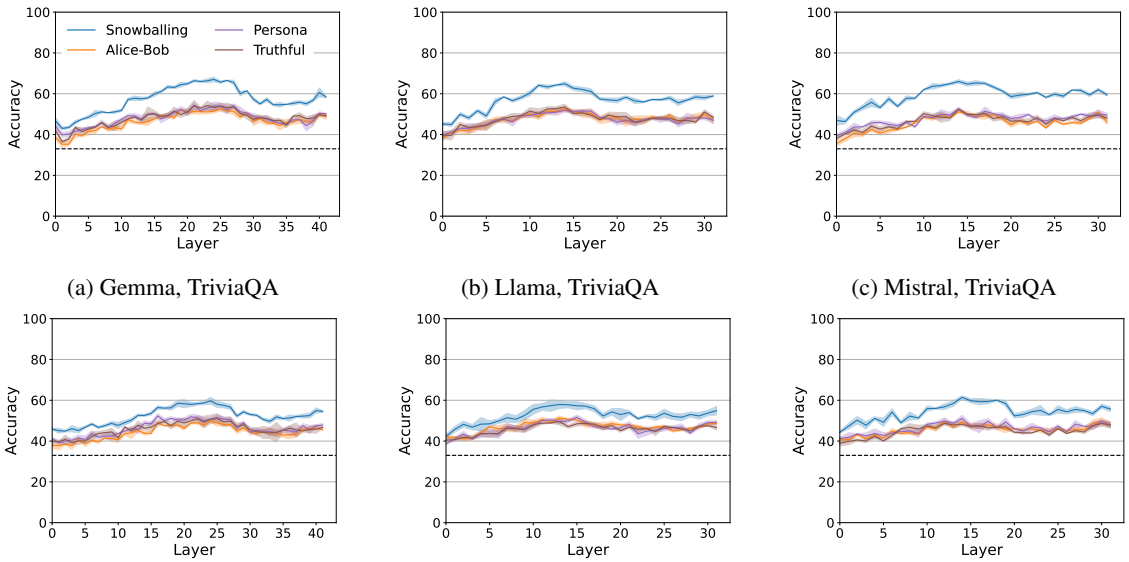
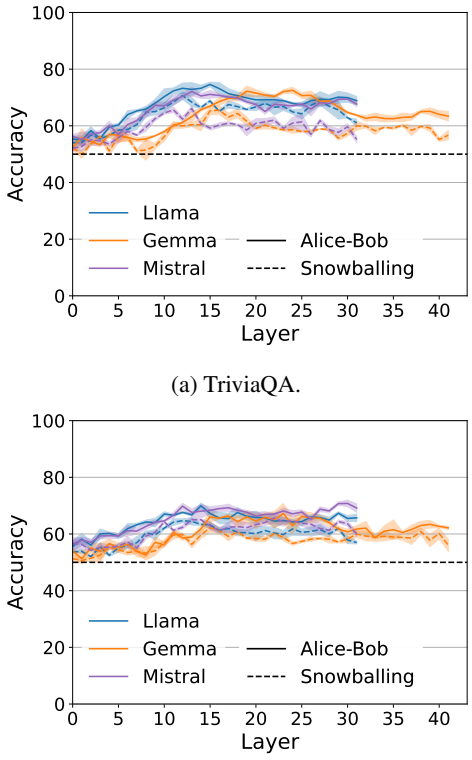


(a) Gemma, TriviaQA (b) Llama, TriviaQA (c) Mistral, TriviaQA

(d) Gemma, Natural Questions (e) Llama, Natural Questions (f) Mistral, Natural Questions

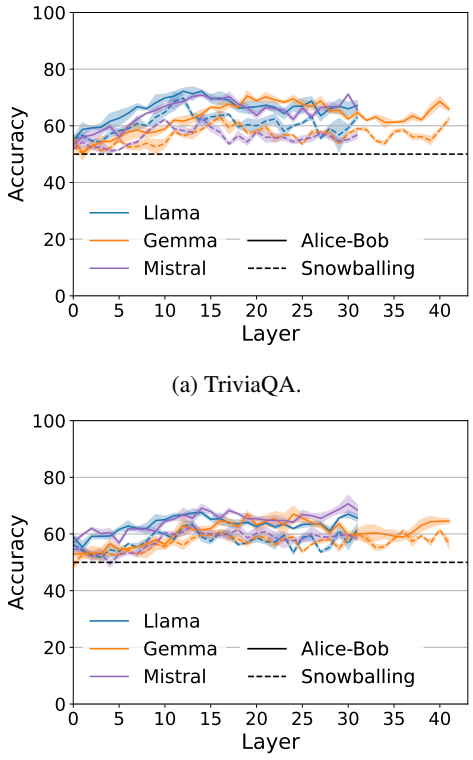
Figure 14: 3-way classification results into (i) hallucinations caused by lack of knowledge (HK^-), (ii) hallucinations caused despite having knowledge (HK^+), and (iii) factually correct examples on the Attention component. We can see that all the detection lines are well above the baseline (dashed line) indicating the possibility of distinguishing HK^+ from HK^-.



(a) TriviaQA.

(b) Natural Questions.

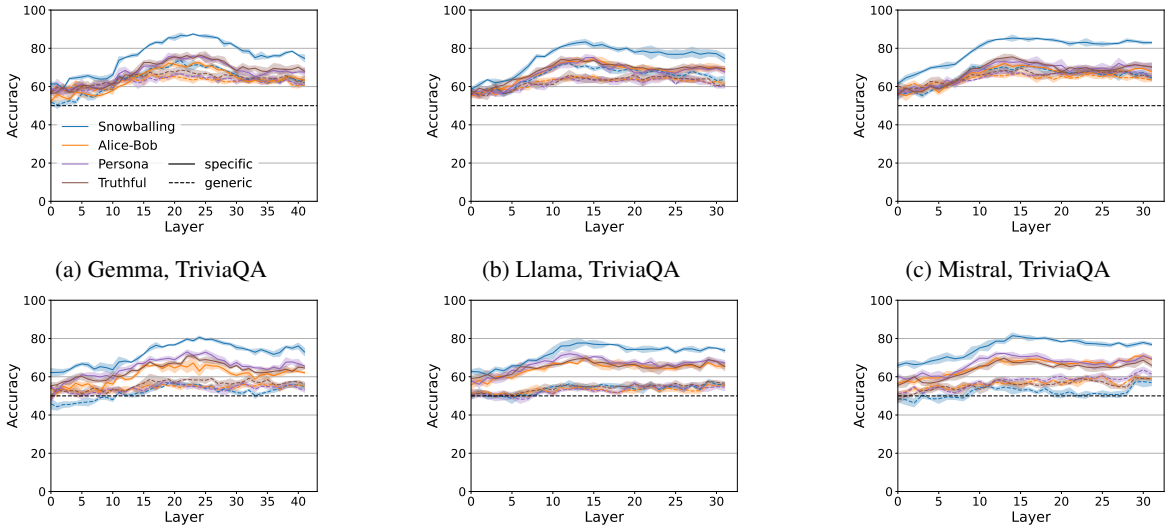
Figure 15: Distinguishing factually correct from HK^+, when training on examples from either a Snowballing setting or an Alice-Bob setting, and testing on the Alice-Bob setting. While the change of setting reduces accuracy, the classifier still performs substantially above a random baseline on MLP component.



(a) TriviaQA.

(b) Natural Questions.

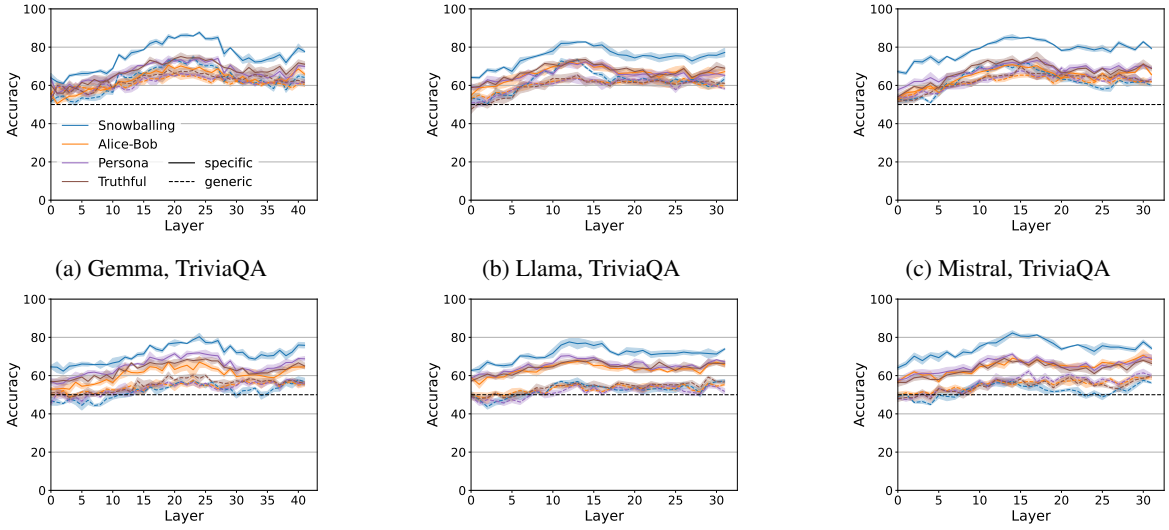
Figure 16: Distinguishing factually correct from HK^+, when training on examples from either a Snowballing setting or an Alice-Bob setting, and testing on the Alice-Bob setting. While the change of setting reduces accuracy, the classifier still performs substantially above a random baseline on Attention component.



(a) Gemma, TriviaQA (b) Llama, TriviaQA (c) Mistral, TriviaQA

(d) Gemma, Natural Questions (e) Llama, Natural Questions (f) Mistral, Natural Questions

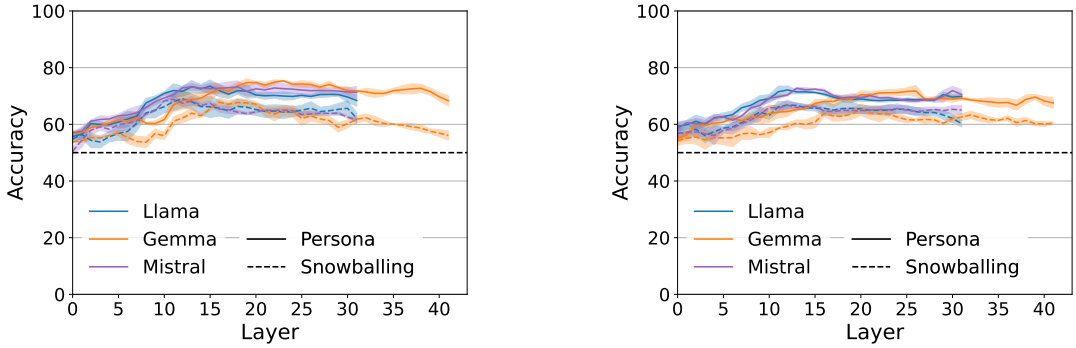
Figure 17: Distinguishing factually correct from HK^+ using classifiers trained on generic vs. model-specific datasets on MLP component. We can see that specific dataset accuracy is significantly higher than generic dataset.



(a) Gemma, TriviaQA (b) Llama, TriviaQA (c) Mistral, TriviaQA

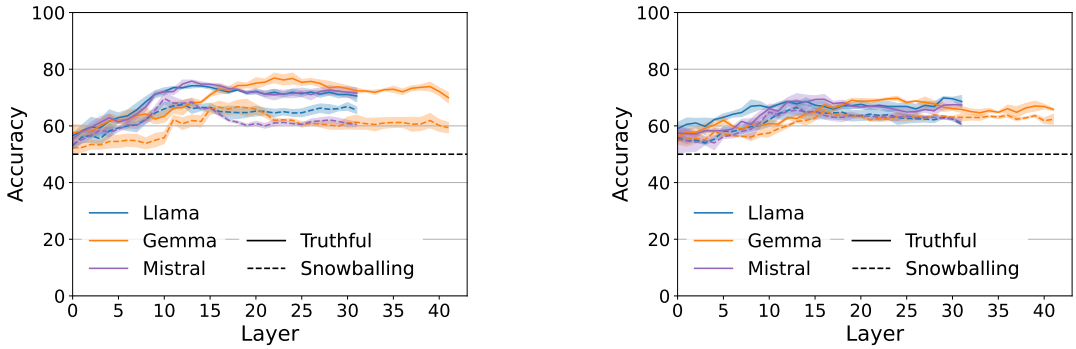
(d) Gemma, Natural Questions (e) Llama, Natural Questions (f) Mistral, Natural Questions

Figure 18: Distinguishing factually correct from HK^+ using classifiers trained on generic vs. model-specific datasets on Attention component. We can see that specific dataset accuracy is significantly higher than generic dataset.



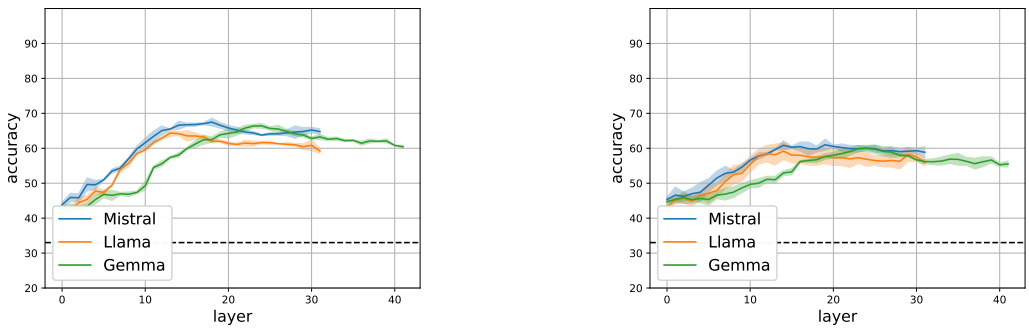
(a) TriviaQA. (b) Natural Questions.

Figure 19: Distinguishing factually correct from HK^+, when training on examples from either a Snowballing setting or Persona setting, and testing on the Persona setting. While the change of setting reduces accuracy, the classifier still performs substantially above a random baseline.



(a) TriviaQA. (b) Natural Questions.

Figure 20: Distinguishing factually correct from HK^+, when training on examples from either a Snowballing setting or a Truthful setting, and testing on the Truthful setting. While the change of setting reduces accuracy, the classifier still performs substantially above a random baseline.



(a) TriviaQA, 5-Snowballing-shots. (b) Natural Questions, 5-Snowballing-shots.

Figure 21: 3-way classification results into (i) hallucinations caused by lack of knowledge (HK^-), (ii) hallucinations caused despite having knowledge (HK^+), and (iii) factually correct examples. We show results using a 5-Snowballing setting.